

1. Chi-Squared Goodness of Fit Test: (40 pts) Gender and number of failures with expected frequencies being equal. (HINT: Sum the number of women who fail and the number of men who fail to form your contingency table). You may choose either dataset, or the merged dataset. If you choose the merged, only use one column (either the .y failures or the .x failures).

Research Hypotheses: There is a significant association between gender and the number of past class failures.

Null Hypotheses: There is no significant association between gender and the number of past class failures.

Alpha: The significance level is set at 0.05, indicating a 5% chance of rejecting the null hypothesis when it is actually true.

Rejection Criteria: We reject the null hypothesis if the Chi-Squared Goodness of Fit Test produces a p-value less than or equal to the chosen significance level ($\alpha = 0.05$). This implies that there is compelling data to relate the gender of students and the number of past class failures. In the event that the p-value exceeds 0.05, the null hypothesis is not rejected, indicating insufficient data to draw any conclusions.

Outcome and reason for that outcome: The outcome of the Chi-Squared Goodness of Fit Test was that it resulted in a chi-squared statistic of 5.4507 with 3 degrees of freedom and a p-value of 0.1416. The calculated p-value (0.1416) is greater than the chosen significance level of 0.05. This indicates that there is insufficient evidence to reject the null hypothesis.

Conclusion: The Chi-Squared Goodness of Fit Test result indicates that we are unable to reject the null hypothesis. Thus, the data does not support the conclusion that there is a meaningful relationship between the number of past class failures and student gender.

2. ANOVA: (40 pts) Choose 2 factors that you will test to see if they are predictive of G3 grades. You can use either data set or the merged dataset. (For example: weekday alcohol consumption and number of failures). Be sure to test for assumptions and test

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both the additive and interactive models. For any statistically significant results, perform a *post hoc* TukeyHSD and report your results as shown in class.

Research Hypotheses: There is a significant relationship between weekday alcohol consumption and the number of failures among secondary education students.

Alternative Hypothesis: The higher the level of weekly alcohol consumption, the higher the chance of that student failing.

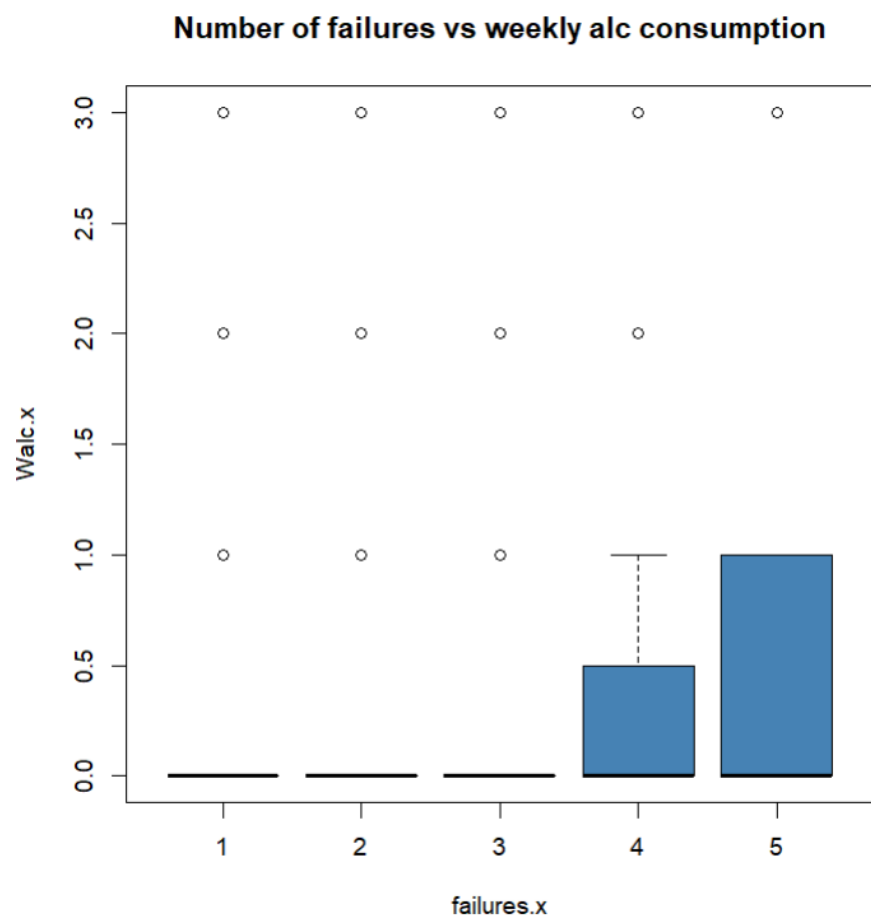
Null Hypothesis: There is no association between the level of weekly alcohol consumption and the likelihood of student failure.

According to my alternate hypothesis, there is a positive correlation between weekday alcohol intake and student failure, meaning that the likelihood of student failure increases with alcohol consumption. Conversely, the null hypothesis suggests that any observed link is the result of random variation and that there isn't a relationship at all.

Alpha: The significance level for this ANOVA test is set at 0.05, indicating a 5% chance of rejecting the null hypothesis when it is actually true.

Rejection Criteria: With a significance level of 0.05, if the p-value obtained from the ANOVA test is less than or equal to 0.05, we reject the null hypothesis. Otherwise, if the p-value is greater than 0.05, we fail to reject the null hypothesis.

Outcome and reason for that outcome: The number of failures is statistically significantly impacted by the Walc.x variable (weekday alcohol intake) ($p < 0.05$). This suggests that the number of failures varies substantially depending on the amount of alcohol consumed during the workday. According to the significant p-value (0.0252), there appears to be a difference in the mean number of failures between at least one level of alcohol intake over the week and the other levels. However, post hoc tests like Tukey's HSD test would be required to see which particular levels of weekday alcohol use differ in terms of failures.



```
> # Run one-way ANOVA
> dataMerged_1w <- aov(failures.x ~ Walc.x, data = dataMerged)
> summary(dataMerged_1w)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Walc.x	4	5.88	1.4702	2.815	0.0252 *
Residuals	377	196.87	0.5222		

```
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
Tukey multiple comparisons of means
95% family-wise confidence level
```

```
Fit: aov(formula = failures.x ~ walc.x, data = dataMerged)
```

```
$walc.x
```

		diff	lwr	upr	p adj
2-1	0.076873489	-0.19336502	0.3471120	0.9363742	
3-1	0.070836337	-0.20933287	0.3510055	0.9579350	
4-1	0.148687215	-0.18087010	0.4782445	0.7298854	
5-1	0.498224252	0.08327797	0.9131705	0.0095836	
3-2	-0.006037152	-0.31873423	0.3066599	0.9999982	
4-2	0.071813725	-0.28580716	0.4294346	0.9818472	
5-2	0.421350763	-0.01621648	0.8589180	0.0654111	
4-3	0.077850877	-0.28733214	0.4430339	0.9773469	
5-3	0.427387914	-0.01638120	0.8711570	0.0653430	
5-4	0.349537037	-0.12695459	0.8260287	0.2627934	

Tukey HSD test outcome: The comparison between level 5 of weekday alcohol consumption and level 1 has a statistically significant difference in mean number of failures ($p \text{ adj} = 0.00958$). This suggests that students with level 5 weekday alcohol consumption have a significantly different mean number of failures compared to those with level 1 weekday alcohol consumption.

Conclusion:

In conclusion, the results of Tukey's HSD test show that while level 5 of weekday alcohol consumption is linked to a significantly different mean number of failures than level 1, there are no significant differences in mean number of failures between the other levels of weekday alcohol consumption. The alternative hypothesis proves true. The higher the level of weekly alcohol consumption, the higher the chance of that student failing.

3. Choose an additional test to perform on this dataset. It could be a t-test (remember that the G1, G2 and G3 grades for mat or por are not independent, but G1 for por is independent of G1 for mat etc.), a correlation analysis or just another comparison using one of the tests above that you are curious about. (40 pts)

I am comparing the mean workday alcohol consumption of math and Portuguese students using an independent samples t-test.

Research Hypotheses: There is a significant difference between the mean workday alcohol consumption of math and Portuguese students.

Null Hypotheses: There is no significant difference between the mean workday alcohol consumption of math and Portuguese students.

Alpha: The significance level for this t-test is set at 0.05, indicating a 5% chance of rejecting the null hypothesis when it is actually true.

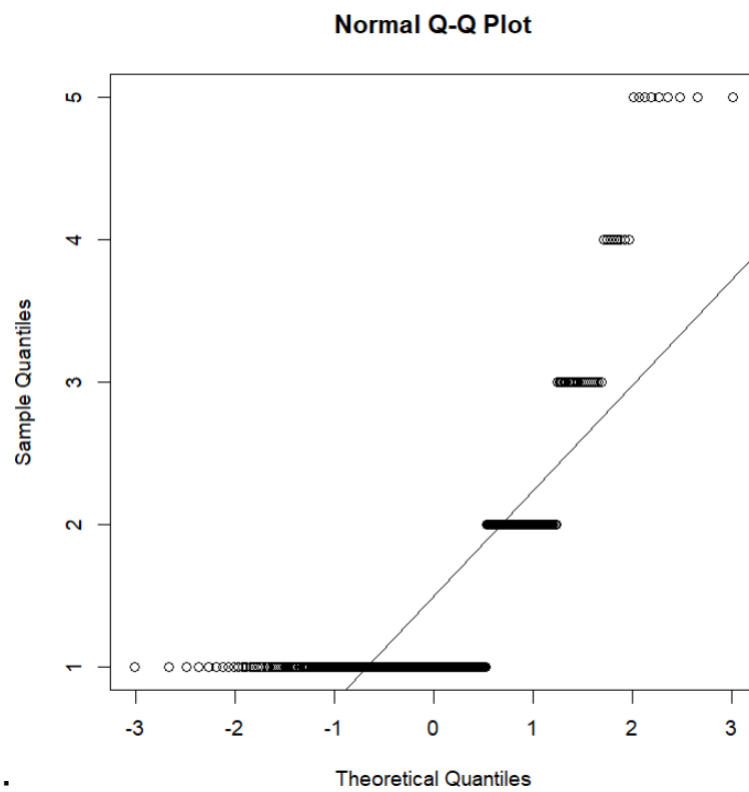
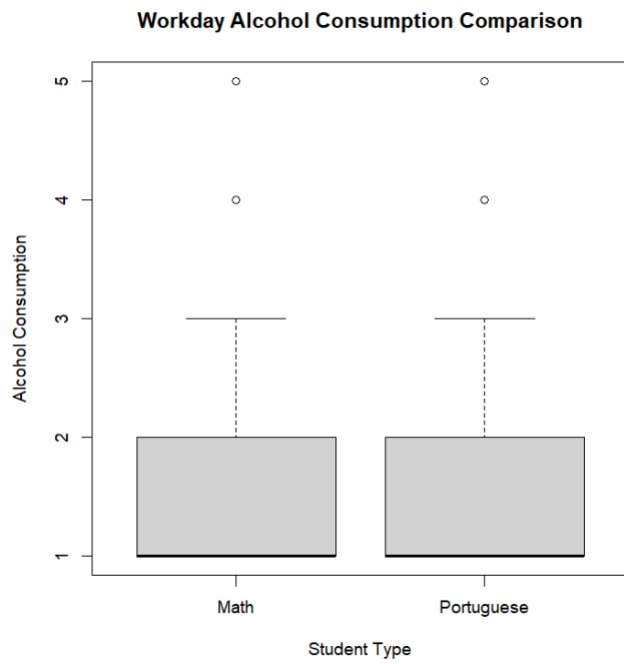
Rejection Criteria: If the p-value is less than or equal to the significance level, then reject the null hypothesis. This means that there is evidence to conclude that there is a significant difference between the mean workday alcohol consumption of math and Portuguese students. If the p-value is greater than the significance level, then fail to reject the null hypothesis. This means that there isn't enough evidence to conclude that there is a significant difference between the mean workday alcohol consumption of math and Portuguese.

Testing assumptions: Used Shapiro test to check for normality.

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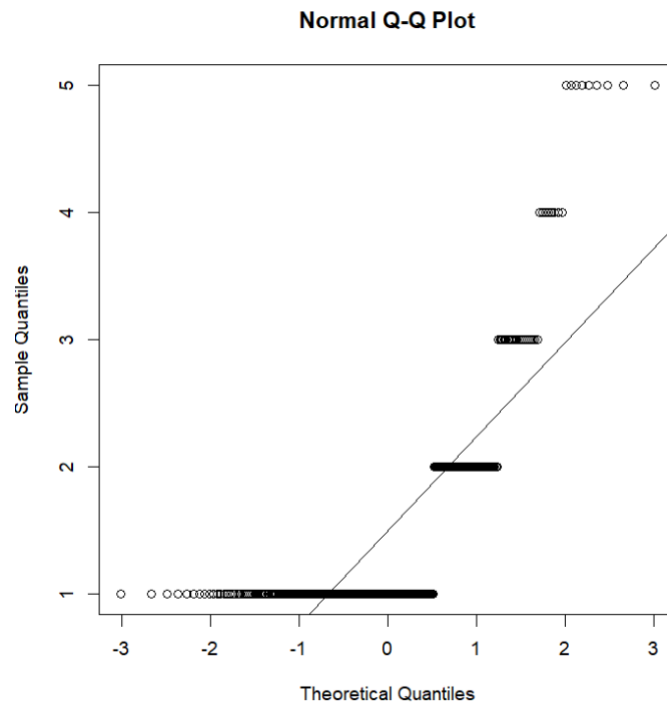


Math alcohol consumption:

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Portuguese alcohol consumption:

```
> #display the central tendency for each year
> summary(alc_math)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1.000  1.000   1.000  1.474  2.000   5.000
> summary(alc_portuguese)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1.000  1.000   1.000  1.476  2.000   5.000
> mean_cl_normal(alc_math)
      y      ymin      ymax
1 1.473822 1.384667 1.562977
> mean_cl_normal(alc_portuguese)
      y      ymin      ymax
1 1.47644 1.387278 1.565602
> # Perform the variance test
> variance_test <- var.test(alc_math, alc_portuguese)
> # Print the result
> print(variance_test)

      F test to compare two variances

data:  alc_math and alc_portuguese
F = 0.99983, num df = 381, denom df = 381, p-value = 0.9987
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.8176744 1.2225742
sample estimates:
ratio of variances
 0.9998338

> # Perform the independent samples t-test
> result <- t.test(alc_math, alc_portuguese, paired = FALSE, var.equal = TRUE, alternative = "two.sided", )
> # Print the result
> print(result)

      Two Sample t-test

data:  alc_math and alc_portuguese
t = -0.040822, df = 762, p-value = 0.9674
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.1285062  0.1232706
sample estimates:
mean of x mean of y
1.473822  1.476440
```

Outcome and reason for that outcome: The p-value obtained from the t-test is 0.9674, which is greater than the significance level of 0.05. Therefore, we fail to reject the null hypothesis.

Conclusion: Since we fail to reject the null hypothesis, it means that there isn't enough evidence to conclude that there is a significant difference between the mean workday alcohol consumption of math and Portuguese students. This conclusion is supported by the small t-value (-0.040822) and the narrow confidence interval for the difference in means (-0.1285062 to 0.1232706), which both indicate that there is very little to no difference between the two groups in terms of alcohol consumption on workdays.

Additionally, the variance test (Levene's test) indicated that the variances of the two groups are not significantly different (p-value = 0.9987), and the var.equal parameter was set to TRUE in the t-test, assuming equal variances between groups.

Therefore, based on the given data and analysis, there is no significant difference in the mean workday alcohol consumption between math and Portuguese students.